From Foundation Models to Clinical Prediction: scGPT → Path-GPTOmic

Understanding how foundation models are extended in single-cell biology

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Why Do We Need Foundation Models in Biology?

- Single-cell RNA-seq → high-dimensional & noisy
- Millions of cells, thousands of genes
- Traditional models:
  a. Task-specific
  b. Limited generalization

Need: General-purpose models for biology

scGPT: A Foundation Model for Single-Cell Data

- Transformer-based model
- Treats genes like tokens
- Pretrained on millions of cells
- Learns biological representations

How scGPT Work

Input (Gene Expression) → Embedding → Transformer → Task Head

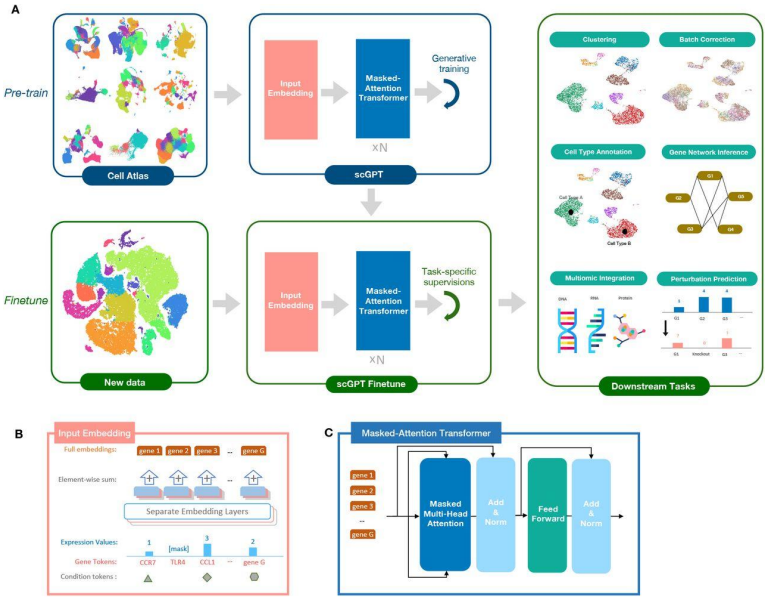

Figure 1: Model Schematic.

What scGPT Cannot Do Well

- Designed for single-cell data
- Limited for:
  - Clinical tasks (e.g., survival prediction)
  - Multi-modal data (images + genomics)

Needs adaptation for real-world applications

Path-GPTOmic: Extending scGPT

- Uses scGPT embeddings
- Targets:
  - Survival prediction
  - Multi-modal learning
- Combines:
  - Gene expression
  - Pathology images
- Goal:
  - Bring foundation models into clinical applications

How It Builds on scGPT

- Takes pretrained scGPT representations
- Adapts them to:
  - Bulk RNA-seq
  - Multi-modal inputs
- Introduces:
  - Balanced learning framework
  - Gradient modulation
- scGPT = backbone
- Path-GPTOmic = extension

Path-GPTOmic Pipeline

- Extract embeddings from scGPT
- Combine with image features
- Train model for survival prediction
- Key innovation:
  - Balancing different modalities
  - Improving robustness

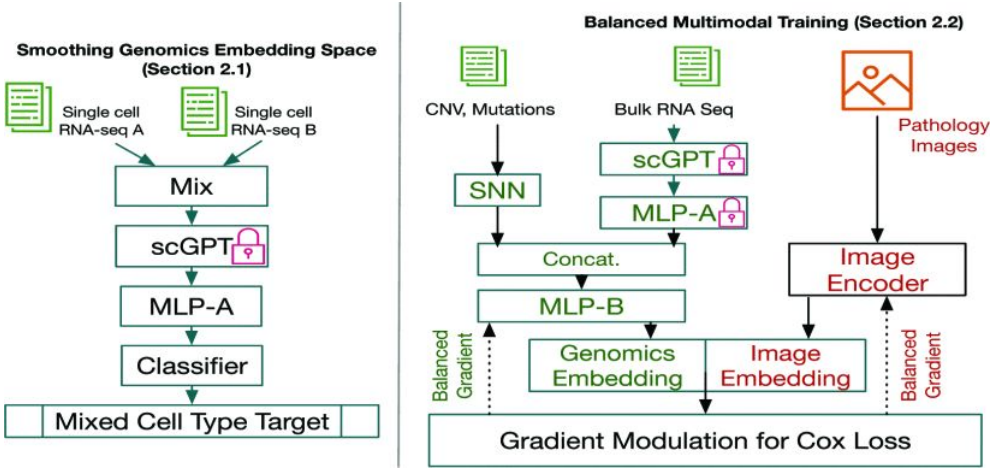

Figure 2: Illustrating Path-GPTOmic pipeline.

Performance Improvements

- Better survival prediction than:
  - Traditional ML models
  - Other deep learning methods
- Works on real datasets (e.g., TCGA)
- Shows practical value of scGPT-based features

Method	C-Index
Histology CNN [3]	0.671 ± 0.023
Histology T2T-ViT [15]	0.683 ± 0.023
Genomic SNN [3]	0.684 ± 0.025
Pathomic Fusion [3] (CNN + SNN)	0.719 ± 0.031
Pathomic Fusion [3] (T2T-ViT [15] + SNN)	0.727 ± 0.033
PathOmics [4]	0.736 ± 0.024
Ours	**0.754 ± 0.030**

Table 1: Comparison of C-Index performance on the TCGA-KIRC dataset ($p < 0.05$).

Why Path-GPTOmic Cites scGPT

- scGPT provides:
  - Pretrained biological knowledge
  - General-purpose embeddings
- Path-GPTOmic adds:
  - Task-specific adaptation
  - Multi-modal learning
  - Clinical application
- Relationship:
  - Extension + Application

Future of Biomedical Research

- Foundation models are emerging in biology
- scGPT = general model
- Path-GPTOmic = applied extension
- Big idea:
  - Pretrained models can accelerate biomedical research

Following [8], we represent W as a combination of two blocks W^G and W^P, and rewrite the equation as:

$$\theta_k = W^G \cdot \mathbf{G_k} + W^P \cdot \mathbf{P_k} + \mathbf{b}. \tag{1}$$

To predict survival outcome, Cox partial log-likelihood [14, 3] is used as cost function:

$$L_{Cox} = \sum_{C(k)=1} (\theta_k - \log \sum_{R(t_k)} \exp(\theta_k)), \tag{2}$$

where $C(k) = 1$ represents the uncensored events, and $R(t_k)$ is the risk set at time t_k. The gradient is:

$$\frac{\partial L_{Cox}}{\partial \theta_k} = \sum_{C(k)=1} (1 - \frac{\exp(\theta_k)}{\sum_{R(t_k)} \exp(\theta_k)}) \tag{3}$$

$$= \sum_{C(k)=1} (1 - \frac{e^{(W^G G_k + W^P P_k + b)}}{\sum_{R(t_k)} e^{(W^G G_k + W^P P_k + b)}}). \tag{4}$$

Balanced Training. To alleviate the imbalance training problem, we evaluate the contribution discrepancy ratio ρ^G, ρ^P for Cox partial log-likelihood loss, as:

$$\rho^G = \left(\frac{(W^G \mathbf{G_k} + \frac{\mathbf{b}}{\mathbf{2}})}{\sum_{R(t_k)} e^{(W^G \mathbf{G_k} + \frac{\mathbf{b}}{\mathbf{2}})}}\right) / \left(\frac{(W^P \mathbf{P_k} + \frac{\mathbf{b}}{\mathbf{2}})}{\sum_{R(t_k)} e^{(W^P \mathbf{P_k} + \frac{\mathbf{b}}{\mathbf{2}})}}\right), \quad (5)$$

$$\rho^P = 1/\rho^G. \quad (6)$$

Formally, inspired by [8], we modulate the gradient by estimating the contributions of the two modalities. We update the model parameters in each iteration t, as:

$$\phi^G_{t+1} = \phi^G_t - \eta \cdot \min(1 - \tanh(\rho^G - 1), 1) \cdot g(\phi^G_t), \quad (7)$$

$$\phi^P_{t+1} = \phi^P_t - \eta \cdot \min(1 - \tanh(\rho^P - 1), 1) \cdot g(\phi^P_t), \quad (8)$$

where ϕ^G_{t+1} and $g(\phi^G_t)$ represent the parameters and gradient for the MLP in the genomics network, ϕ^P_{t+1} and $g(\phi^P_t)$ represent the parameters and gradient for the pathology image network, and η is the learning rate. In this way, the learning rate of the modality with higher ρ is suppressed.